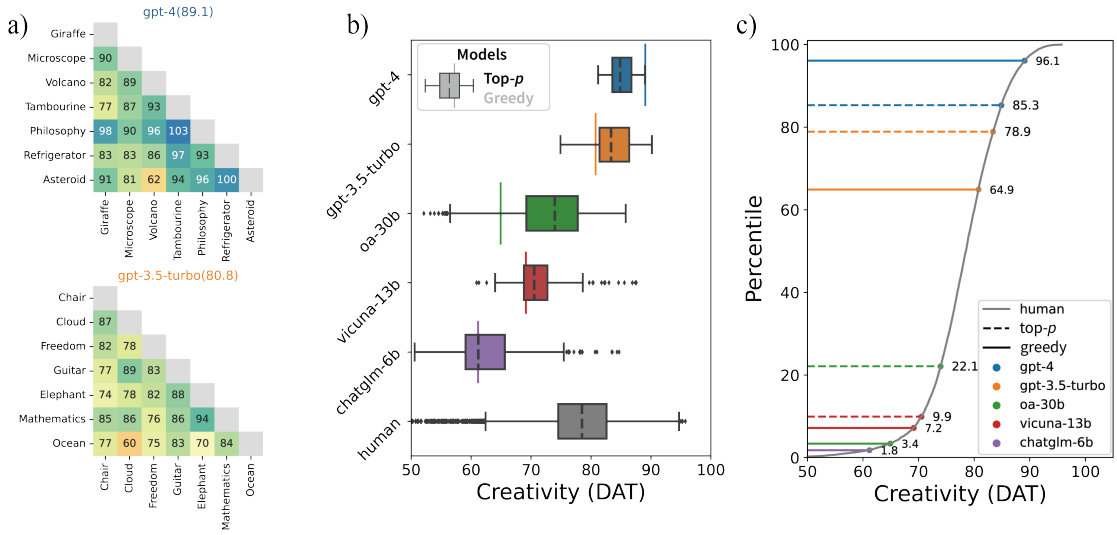


Figure 3: The DAT for humans and models. (a) The distance matrix of words generated by GPT-4 and GPT-3.5-turbo. The average distance is defined as the DAT. (b) The DAT of models and human. (c) The percentile of models' DAT against human results.

## 4 Result

The DAT results are shown in Figure 3. Using greedy search, GPT-4 achieves the highest DAT of 89.1, surpassing 96.1% of humans, and GPT-3.5-Turbo attains a DAT of 80.8 that is above the average human-level (Figure 3c). Other models perform less well with lower DAT, which are roughly proportional to the size of models. When using top-$p$ sampling, models other than GPT-4 are capable of getting the DAT much higher than greedy search, but they also become unstable that probably generate answers with low DAT scores (Figure 3b).

We also report the relation between the DAT and surprisal in Figure 4.[6] Theoretically, surprisal is corresponding to the novelty that is an element of creativity, and the original DAT metric including word frequency effect is valid for human as well (Olson et al., 2021). But as mentioned in Section 3.3, word frequency might be a confounding variable when calculating semantic distance. Indeed, we find two variables highly relevant for human and random baselines (also see Appendix B). For models, the results of top-$p$ sampling have homogeneous slopes with two baselines, but their intercepts and surprisal distributions are different. GPT-4 and GPT-3.5-Turbo exceed the average human DAT under the same surprisal, while other models fall short. Despite Vicuna-13B and Chatglm-6B have similar distributions of surprisal, the former generates words more divergently. Oasst-Llama-30B defeats Vicuna-13B on the DAT, but this might be explained by the capacity or tendency to generate rarer words. To clarify this effect, we control the surprisal for DAT in Appendix D. The results are similar that GPT-4 and GPT-3.5-Turbo outperform average human performance, but the superiority of GPT-4 is attenuated.

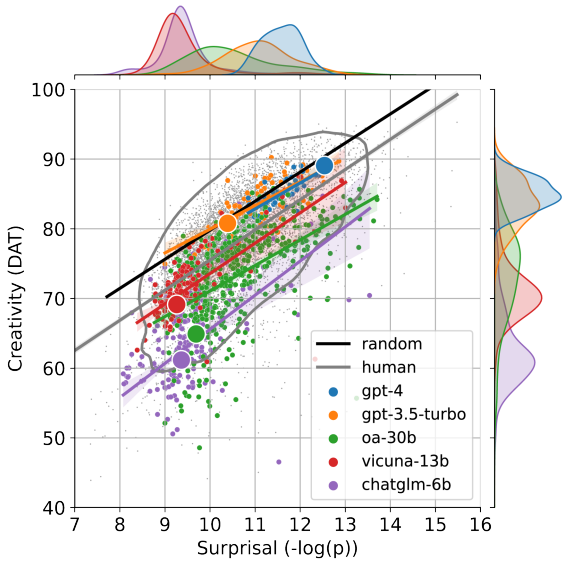


Figure 4: Relationship between the DAT and surprisal. Rimmed points show the results of greedy search. The contour indicates the 95% confidence interval of humans.

[6]The DAT and surprisal for more LLMs are shown in Appendix C

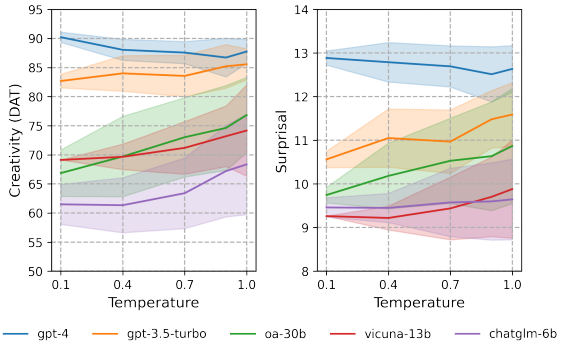


Figure 5: Effect of temperature tuning. The bands indicate the standard deviations.

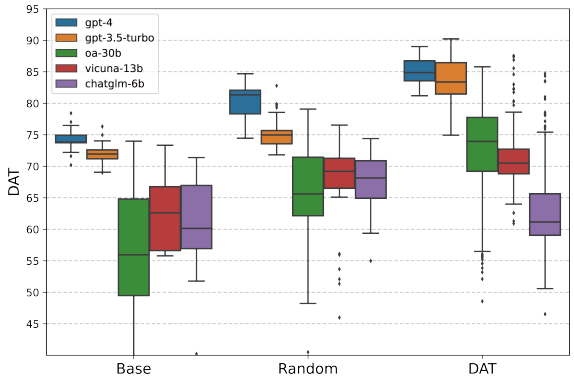


Figure 6: The DAT scores of 3 conditions.

We further investigate the effect of temperature (Figure 5). Temperature is a widely deployed parameter for creative or original generation in practice that high temperature skews the distribution shape towards low-probability tokens (Ackley et al., 1985; Fan et al., 2018). We vary the temperature from 0.1 to 1 and found its positive effect for models except GPT-4. However, the effect is limited, and high temperature will also bring instability and produce invalid answers. As for GPT-4, high-probability tokens are well aligned with high-quality answers.

In the relationship between the DAT and surprisal, we find a naive algorithm that samples nouns from Wordnet can outperform the majority of humans and models (Figure 4). It is because the algorithm has no constrains of the language distribution, which also means it can barely accomplish general language tasks. Although LLMs have exhibited striking mastery of natural language, we wonder whether they process the semantics differently with humans and the DAT test is accordingly invalid as well. Thus, we compare the DAT with Base and Random conditions (figure 6). We show that if not instructed, LLMs tend to produce more related words. When instructed with the prompts of Random and the DAT, LLMs can modulate the language distributions to be more divergent.

These results indicate that LLMs have the potential to generate divergent content with instruction, but with the flaw of not inhibiting common words. Stochastic decoding strategy is helpful for promoting remote association, but since the creative and nonsense content are both infrequent, it cannot accurately produce high-quality content.[7] However, advanced LLMs show the implicit control of modulating the probability distribution and stably generating divergent answers. Stochastic decoding strategy may even degrade performances for introduced randomness.

## 5 Conclusion

In this work, we provide a creativity evaluation using a divergent semantic task. This task reveals distinguishable results across various LLMs and decoding strategies. We find GPT-4 demonstrates advanced human-level creativity stably, while GPT-3.5-turbo exceeds the average human level. For decoding methods, stochastic decoding strategy is effective but not enough for creative generation.

## Limitation

Creativity is a deeply debated concept. We selectively evaluate the "little-C" (creativity in everyday life) that is a general capacity, instead of the "big-C" (marvelous creative product) which is rare even for human. Measuring creativity is also controversial that requires evaluations from multiple perspectives, principles, and analysis. Thus, the results of this study cannot be directly generalized to all language generation tasks. We also limited the range of this study within self-contained creativity, whereas another crucial aspect of AI's creativity is human-AI co-creation. We leave these for future work.

## Acknowledgement

We thank Mark Sun and the anonymous reviews for their thoughtful helps and suggestions. This work was supported by STI2030-Major Projects 2021ZD0204105 and Fundamental Research Funds for the Central Universities 226-2023-00091.

[7] Previous psychological researches reported similar result that mild imperfection of attention is related to higher creativity (Abraham, 2014).

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